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20AM5T01 - AUTOMATA THEORY				

COURSE OUTCOMES:

At the end of the course, student will be able to

CO1: Able to understand the concept of abstract machines and their power to recognize the languages(K1)
CO2: Able to employ finite state machines for modelling and solving computing problems. (K2)
CO3: Able to design context free grammars for formal languages (K2)
CO4: Able to distinguish between decidability and undecidability (K2)
CO5: Able to gain proficiency with mathematical tools and formal methods (K2)

UNIT - I :

Introduction to Finite Automata: Structural Representations, Automata and Complex- ity, the Central Concepts of Automata Theory – Alphabets, Strings, Languages, Problems.

Deterministic Finite Automata: Definition of DFA, How A DFA Process Strings, The language of DFA.

Nondeterministic Finite Automata: Formal Definition, an application, Finite Automata with Epsilon-Transitions Conversion of NFA with C-transitions to NFA without Ctransi- tions. Conversion of NFA to DFA.

UNIT - II :

Regular Expressions: Finite Automata and Regular Expressions, Applications of Regular Expressions, Algebraic Laws for Regular Expressions, Conversion of Finite Automata to Regular Expressions.

Pumping Lemma for Regular Languages: Statement of the pumping lemma, Applications of the Pumping Lemma.

UNIT - III :

Context-Free Grammars: Definition of Context-Free Grammars, Derivations Using a Grammar, Leftmost and Rightmost Derivations, the Language of a Grammar, Sentential Forms, Parse Tress, Applications of Context-Free Grammars, Ambiguity in Grammars and Languages.

Push Down Automata: Definition of the Pushdown Automaton, the Languages of a PDA, Equivalence of PDA's and CFG's, Acceptance by final state, Acceptance by empty stack, Deterministic Pushdown Automata.

UNIT - IV :

Normal Forms for Context- Free Grammars: Eliminating useless symbols, Eliminating CProductions. Chomsky Normal form, Griebech Normal form.

Pumping Lemma for Context-Free Languages: Statement of pumping lemma, Applica- tions

Turing Machines: Introduction to Turing Machine, Formal Description, Instanta- neous description, The language of a Turing machine.

UNIT - V :

Types of Turing machine: Turing machines and halting

Undecidability: Undecidability, A Language that is Not Recursively Enumerable, An Undecidable Problem That is RE, Recursive languages, Properties of recursive languages, Post's Correspondence Problem, Counter machines.

Text Books:

1. Introduction to Automata Theory, Languages, and Computation, 3nd Edition, John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, Pearson Education.
2. Theory of Computer Science – Automata languages and computation, Mishra and Chandrasekaran, 2nd edition, PHI.

Reference Books:

1. Introduction to Languages and The Theory of Computation, John C Martin, TMH.
2. Introduction to Computer Theory, Daniel I.A. Cohen, John Wiley.
3. A Text book on Automata Theory, P. K. Srimani, Nasir S. F. B, Cambridge University Press.
4. Introduction to the Theory of Computation, Michael Sipser, 3rd edition, Cengage Learning.
5. Introduction to Formal languages Automata Theory and Computation Kamala Krithivasan, Rama R, Pearson.